Predictive Phase Continuation:
A Hierarchical Benchmark from Solvable Models to Realistic Learning Systems

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Abstract

Solvable learning models expose sharp changes in recovery, specialization, and collective behavior, but their assumptions are far from those of trained neural systems. We ask which phase structure and which order-parameter semantics survive as those assumptions are relaxed. We introduce a hierarchical theory-to-reality taxonomy over data or environment, model or interaction, objective or feedback, learning dynamics, scale or resources, and lifecycle or population, together with *predictive phase continuation.* A solvable or oracle anchor calibrates a restricted surrogate that must predict a boundary, transition diagnostic, and intervention response on an untouched deformation. Prediction is compared with simple transport baselines; failure is classified before any additional macrovariable is evaluated.

Transformer experiments are the flagship mechanism study. A broad five-seed screen measures attention, feed-forward, residual, objective, optimizer, scaling, and natural-data coordinates with separate $O(1)$ train, IID-test, and OOD risks. Frozen confirmation experiments then target parameter-matched MLP effects, optimizer contrasts, uncapped data scaling, and byte-disjoint corpus evaluation. Diffusion, reinforcement learning, and multi-agent systems provide canonical validation of the same experimental grammar without assuming common critical exponents. The resulting artifact is a falsifiable benchmark and an evidence ledger, not a claim that every learning curve is a thermodynamic phase transition or that controlled boundaries already transfer to open-weight models.

1 Introduction

Statistical mechanics is most useful for machine learning when it predicts what happens after a simplifying assumption is removed. A phase diagram drawn after the fact can summarize a simulator, but it does not establish that a solvable theory retains explanatory power for a trained system. We therefore study one question:

> Which phase structures and order parameters survive along controlled paths from solvable models to realistic learning systems?

Our answer has three layers. First, a *hierarchical realism taxonomy* states what is changed: data or environment, model or interaction, objective or feedback, optimization dynamics, scale or resources, and lifecycle or population. Second, *predictive phase continuation* states how a change is judged: calibrate at an anchor, predict an untouched deformation, diagnose breakdown, and test a preregistered repair on independent evidence. Third, domain experiments state what was learned. Transformers receive the deepest mechanistic treatment; diffusion, reinforcement learning (RL), and multi-agent systems test external validity of the experimental grammar.

This asymmetry is deliberate. The paper does not propose one universality class for four unrelated dynamical systems. It proposes one evidence protocol for asking whether an order parameter, boundary, or intervention remains predictive. The protocol separates a genuine crossing from a censored threshold, an $O(1)$ risk from an extensive resource, and a five-seed taxonomy screen from a finite-size phase claim.

The contributions are:

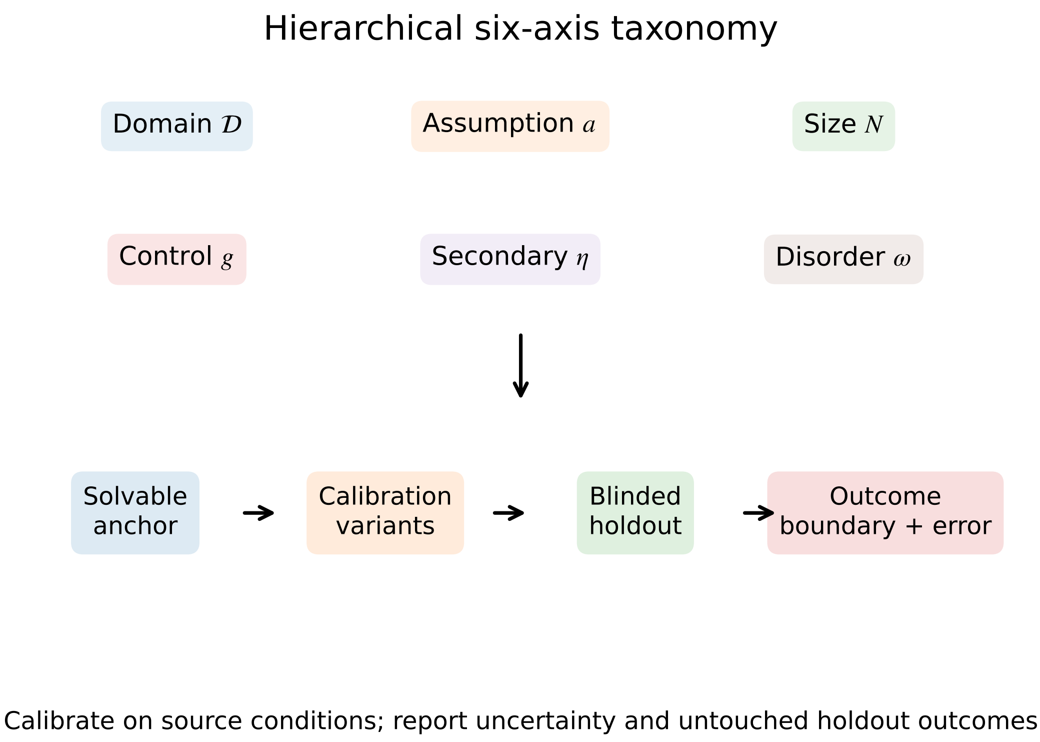

Figure 1: The paper in one figure. A hierarchical realism axis selects a deformation; effective variables are calibrated at an anchor, scored on an untouched holdout, and assigned an outcome class. The diagram separates taxonomy (what changes), prediction (what must be forecast), and evidence (what can be claimed).

1. a factorized design space, an order-parameter dictionary, an outcome vocabulary, and a PhaseCard that make theory-to-reality deformations comparable without erasing domain-specific physics;
2. a held-out prediction test with nontrivial baselines, nested disorder, explicit censoring, and evidence tiers that bound the language allowed for each result;
3. a Transformer flagship study that distinguishes broad exploratory coverage from frozen confirmation of module, optimizer, scaling, and data hypotheses; and
4. canonical diffusion, RL, and multi-agent validations, plus a complete claim-to-artifact ledger and portable Spark-only reproduction workflow.

2 A hierarchical taxonomy of theory-to-reality deformations

The design space is a graph rather than a fully observed Cartesian tensor. A node is a learning system; an edge removes one registered assumption. Three levels prevent axes of different abstraction from being conflated.

Level 0: domain. The domains are Transformer representations, diffusion trajectories, RL feedback dynamics, and multi-agent populations.

Axis	Transformer	Diffusion	RL / RLVR	Multi-agent
Data / environment	synthetic, grammar, web text	mixture, manifold, image data	bandit, MDP, verifier tasks	private evidence, task graph
Model / interaction	QKV, heads, MLP, norm, residual	score model, U-Net, DiT, locality	tabular, MLP, Transformer policy	topology, bandwidth, memory, roles
Objective / feedback	MSE, CE, causal, SFT, preference	ϵ, x_0, v, flow, guidance	gold, proxy, process reward	individual, team, competitive reward
Dynamics / optimizer	SGD-M, AdamW, Muon, SOAP, schedule	optimizer, noise schedule, solver	PPO, off-policy reuse, search	response and memory update
Scale / resources	d, L, T, H, n, C	dimension, resolution, NFE	parameters, horizon, rollouts	agents, degree, rounds
Lifecycle / population	pretrain, LoRA, pruning, RAG, rollouts	distillation, conditioning	SFT–RL, verifier calls	heterogeneous agents, dynamic topology

Table 1: Hierarchical realism taxonomy. Entries are domain-specific subaxes, not claims that every cell is tested in this paper.

Level 1: thermodynamic level. The object of interest is respectively learning, representation, trajectory, or population order. A system may occupy more than one level, but every PhaseCard selects one primary level.

Level 2: six realism axes. Table 1 is the central design taxonomy. Architecture subcomponents such as attention and MLPs live under model/interaction rather than appearing beside data or population as peer axes.

2.1 What is measured

An order parameter must retain meaning under deformation. We group observables by function rather than by domain (Table 2). In particular, $K_{\mathrm{eff}}^{\mathrm{macro}} = \exp H(Z)$ counts effective macrostates, $N_{\mathrm{eff}}^{\mathrm{corr}}$ discounts correlated components, and $r_{\mathrm{eff}}^{\mathrm{spec}} = (\sum_i \lambda_i)^2 / \sum_i \lambda_i^2$ is a spectral effective rank. They are not interchangeable effective numbers.

2.2 Outcomes and PhaseCards

An edge can preserve a phase structure, renormalize its coordinate, split or merge regimes, round a sharp response, insert a new regime, separate information and computation thresholds, become path dependent, or destroy the semantics of the order parameter. These outcomes are hypotheses adjudicated by registered diagnostics; they are not labels assigned from visual smoothness.

The minimal experimental record is

$$\mathcal{P} = (D, L, A, \delta, \omega, \mathbf{N}, \mathbf{g}, \mathbf{m}, \mathbf{R}, H, E), \tag{1}$$

where D is domain, L thermodynamic level, A anchor, δ deformation, ω disorder hierarchy, $\mathbf{N}$ system-size vector, $\mathbf{g}$ controls, $\mathbf{m}$ primary and secondary orders, $\mathbf{R}$ resources, H holdout rule, and E evidence tier. Full PhaseCards are machine-readable; representative cards appear in Section 5.

3 Theory frontier and the unresolved gap

The taxonomy is our organizational proposal, informed by reproducibility frameworks that make model and dataset assumptions explicit [15, 21, 25]. It is not presented as an established consensus.

Family	Meaning	Representative probe	Required null
Recovery	latent truth recovered	counterfactual semantic or oracle probe	shuffled latent labels
Specialization	permutation symmetry broken	head, branch, strategy, or role assignment	permuted components
Locality	range of causal influence	matched site or message perturbation	random-site perturbation
Multiplicity	independent computational units	correlation participation	independent-replica null
Entropy / commitment	macrostate selection	strategy or trajectory entropy	token-level entropy alone
Replica structure	diversity of solutions	functional overlap distribution	seed-matched random solutions
Robustness	quality of the ordered state	OOD risk, gold reward, truth consensus	proxy-only performance
Resource response	marginal value of compute	matched FLOP, NFE, rollout, or message intervention	equal-resource control

Table 2: Order-parameter dictionary. A PhaseCard records the exact estimator, probe distribution, normalization, units, and null.

Anchor setting	Established result	Restrictive assumption	Deformation tested here
Tied attention	positional–semantic boundary	low rank, attention only	trainable block and module screens
Attention-indexed model	Bayes boundary	simplified data and loss	optimizer, objective, natural carrier
Oracle diffusion score	speciation and collapse regimes	oracle score	trajectory and learned-score validation
Proxy-reward model	Goodhart scaling	controlled verifier	KL–noise phase map
Mean-field consensus	field/coupling response	homogeneous agents	population and field holdouts

Table 3: Theory frontier and the assumption removed by the benchmark. The last column states a test, not a guarantee of successful continuation.

For attention, a tied low-rank model exhibits a positional–semantic transition [6], and attention-indexed models extend Bayes predictions to full-width matrices [4]. The open question is therefore not whether simplified attention can transition, but whether its boundary and order semantics survive standard blocks, MLPs, normalization, causal objectives, optimization, and natural carriers.

Diffusion theory distinguishes speciation from later memorized-point collapse [3]; DDPM and DiT provide the relevant learned endpoints [16, 23]. A recent unreviewed preprint reports nearby symmetry-breaking and nonlocality onsets [30], which we treat as a hypothesis rather than settled evidence. In RL, proxy optimization can reduce gold quality [14, 26], so entropy must be read jointly with gold reward and strategy support. Consensus dynamics have a classical foundation [8]; modern MARL and LLM debate add nonstationarity and heterogeneous language agents [9, 20]. The 2026 statistical-physics agent papers we cite are explicitly recent preprints, useful for hypotheses but not a mature theoretical frontier [1, 7].

4 Predictive phase continuation

Let $\mathcal{O}_{\text{cal}}$ be registered calibration observables and $\mathcal{O}_{\text{solv}}(\boldsymbol{\theta})$ a solvable surrogate. We estimate

$$\widehat{\boldsymbol{\theta}}_{\text{eff}} = \arg\min_{\boldsymbol{\theta}} D\Big(\mathcal{O}_{\text{cal}}^{\text{target}}, \mathcal{O}_{\text{solv}}(\boldsymbol{\theta})\Big)\,, \tag{2}$$

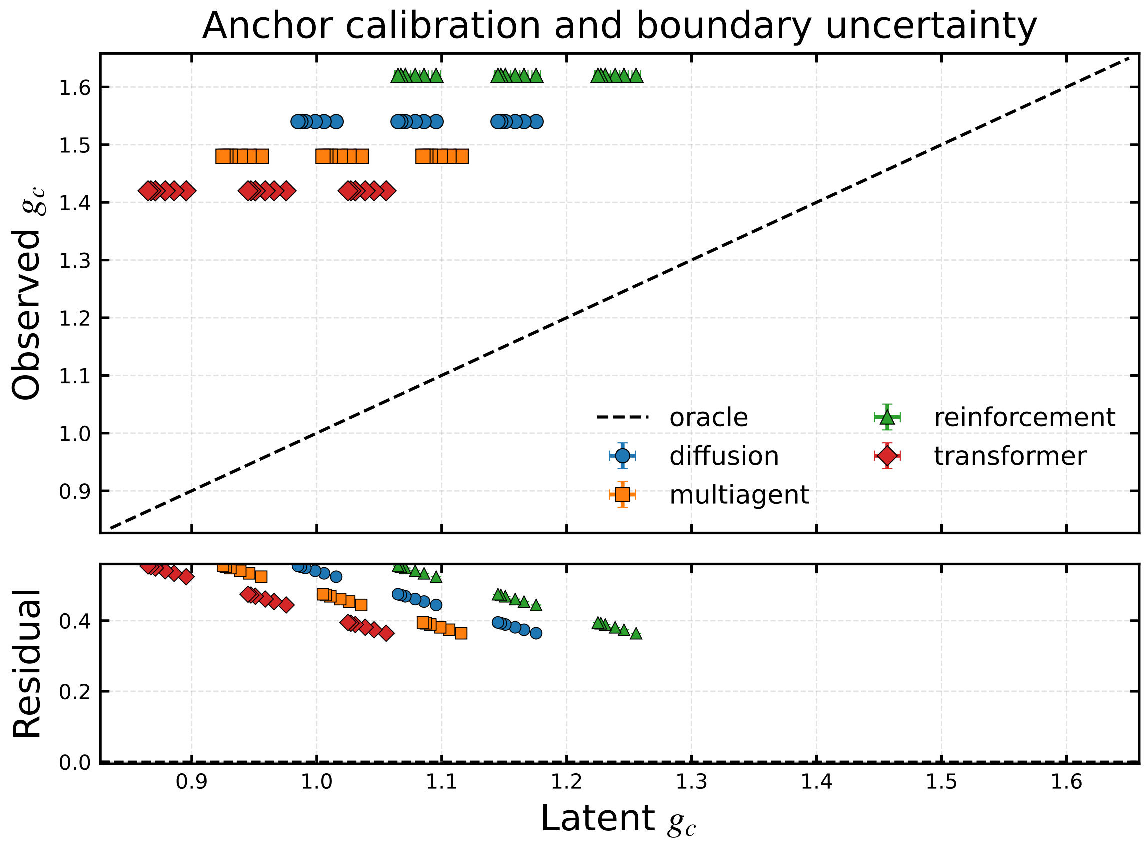

Figure 2: Anchor calibration before continuation. Observed pseudocritical controls are compared with registered latent references, with outer-seed 95% intervals on both axes and a residual panel. This is a calibration diagnostic, not held-out evidence for transport.

then predict an order curve, a boundary $\widehat{g}_c$, transition diagnostics, and an intervention response. The bridge error is evaluated only on withheld quantities,

$$\mathcal{E}_{\text{bridge}} = D\left(\mathcal{O}^{\text{target}}_{\text{holdout}}, \mathcal{O}_{\text{solv}}(\widehat{\boldsymbol{\theta}}_{\text{eff}})\right). \tag{3}$$

Semantic retention $S_{\text{sem}} = I(m; Z)/H(Z)$ is a separate estimand: a small boundary error is meaningless if m no longer tracks the latent variable Z.

Baselines. Every predictive comparison includes a constant predictor, direct transport of the source boundary, a size-only or nearest-calibration predictor, the base surrogate, and the registered augmented surrogate. A diagonal plot without these controls cannot distinguish physical transport from simulator reparameterization.

Breakdown and repair. Two assumptions can interact through

$$\Delta_{ij} g_c = g_c^{ij} - g_c^{i} - g_c^{j} + g_c^{0}. \tag{4}$$

Holdout A is used only to diagnose a missing coordinate. Candidate macrovariables and their selection rule are frozen on a calibration split. Only Holdout B can support a repair claim. When an augmented surrogate was specified before either holdout, we call it a preregistered comparison, not a post-hoc repair.

Tier	Purpose	Sizes	Outer seeds	Allowed language
A	finite-size phase or boundary transport	≥ 6	12–32	phase structure, crossing/crossover, predictive boundary
B	frozen mechanism or endpoint comparison	≥ 3	≥ 5	shift, mechanism candidate, resolved/unresolved contrast
C	natural or open endpoint	≥ 2	≥ 5	measurement transport and bounded external validity

Table 4: Evidence tiers. More seeds do not upgrade an invalid order parameter or a non-independent holdout.

Prediction target. A threshold estimate is recorded as crossed, left-censored, or right-censored. Only crossed estimates enter boundary-error summaries. Censored curves remain visible because the absence of a crossing is often the scientific result.

5 Experimental and statistical protocol

Evidence strength follows Table 4. The controlled predictive benchmark uses 16 independent outer-disorder seeds, four inner stochastic replicates, and 16 additional outer seeds near the frozen boundary window. Inner replicates are never counted as independent disorder. The broad Transformer taxonomy uses exactly five full-pipeline seeds per condition and Student-t intervals; it is a screen. Frozen confirmation uses twelve fresh outer seeds. These designs answer different questions and are never pooled.

Finite-size inference. Finite-size scaling is domain specific [2, 12, 13]. We therefore do not impose $N^{-1/2}$ or N^{-1} on width, trajectory resolution, horizon, and population as universal exponents. Candidate forms are empirical models whose parameters are fit below the largest size and scored prospectively on it. A phase label additionally requires order semantics, sharpening or crossing diagnostics, sensitivity to the minimum size and fitting window, and a registered alternative.

Intensive observables. For a vocabulary of size V, cross entropy is reported as $R_{\mathrm{CE}} = \ell_{\mathrm{CE}}/\log V$; the Brier risk is divided by $1 - 1/V$. We report R_{train}, R_{IID}, R_{OOD}, and $e_{\mathrm{gen}} = R_{\mathrm{IID}} - R_{\mathrm{train}}$ separately. FLOPs, tokens, wall time, width, depth, context, and sample count remain resource coordinates. Causal module effects use raw risk differences and a bounded signed normalization, never division by a vanishing remaining-accuracy scale.

Accounting and preregistration. A condition is a unique PhaseCard; a trained model is one outer seed; an inner replicate is conditional stochasticity; an evaluation is not a new model. Configuration hashes, corpus revisions and hashes, seed values, raw seed metrics, censor flags, cap-hit flags, tokens, estimated FLOPs, and elapsed time are stored before aggregation. Confirmation configurations were frozen after the exploratory audit in the machine-readable claim ledger [22].

6 Transformer flagship study

The Transformer is the flagship because all six realism axes can be varied while retaining a direct task-success probe. The study has two layers: a five-seed atlas that locates candidate effects and a twelve-seed frozen suite that tests only claims selected before the new seeds are observed.

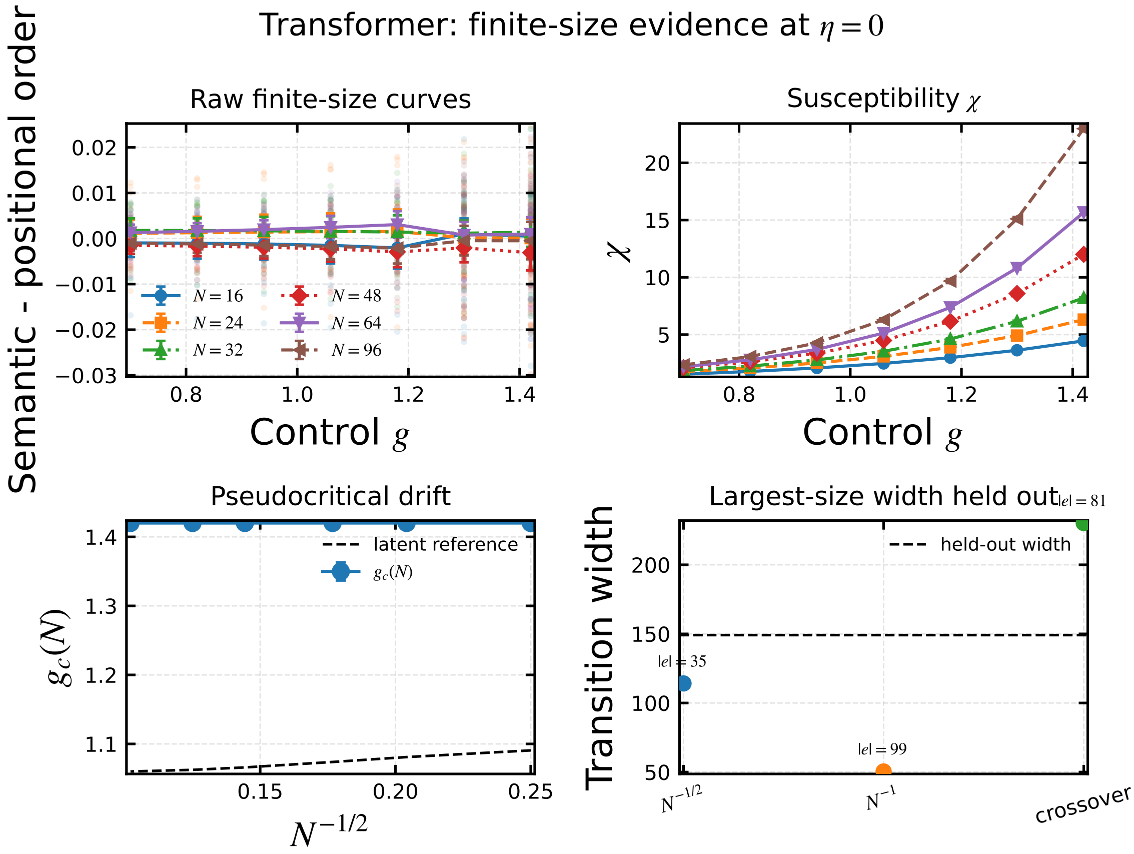

Figure 3: Transformer anchor finite-size evidence. Panels separate the raw order curve, response, pseudocritical drift, and held-out model comparison. Only explicit crossings are boundary estimates; endpoint-only curves are censored.

6.1 Data and attention anchor

The controlled retrieval task contains a latent key–value relation and a positional shortcut. The primary task order is held-out answer accuracy; attention entropy, causal range, and representation overlap are secondary diagnostics. Figure 3 shows the Tier-A controlled anchor with raw disorder points, susceptibility, drift, and largest-size model comparison. Performance-derived $1 - R$ is retained as a learning diagnostic, not silently promoted to a thermodynamic order.

6.2 Feed-forward modules: from screen to matched confirmation

The atlas varies no MLP, linear, ReLU, GELU, GEGLU, and SwiGLU blocks across width and sample coefficient. Its completed five-seed aggregate shows that gated MLPs do not monotonically improve the selected attention-dominant endpoint. This is an MLP-dependent response, not yet evidence of a new phase. The frozen suite compares linear/GELU blocks with gated blocks whose hidden ratio is reduced from 4 to 8/3, matching feed-forward parameter count. It records full, attention-ablated, MLP-ablated, and jointly ablated $O(1)$ risks.

At the largest registered endpoint, the normalized task orders are 0.514 ± 0.007 (linear), 0.521 ± 0.007 (GELU), 0.433 ± 0.016 (GEGLU), and 0.439 ± 0.013 (SwiGLU). The corresponding bounded MLP effects are 0.254 ± 0.080, 0.310 ± 0.092, 0.187 ± 0.049, and 0.183 ± 0.026. These values are generated directly from the frozen 12-seed aggregate rather than transcribed by hand.

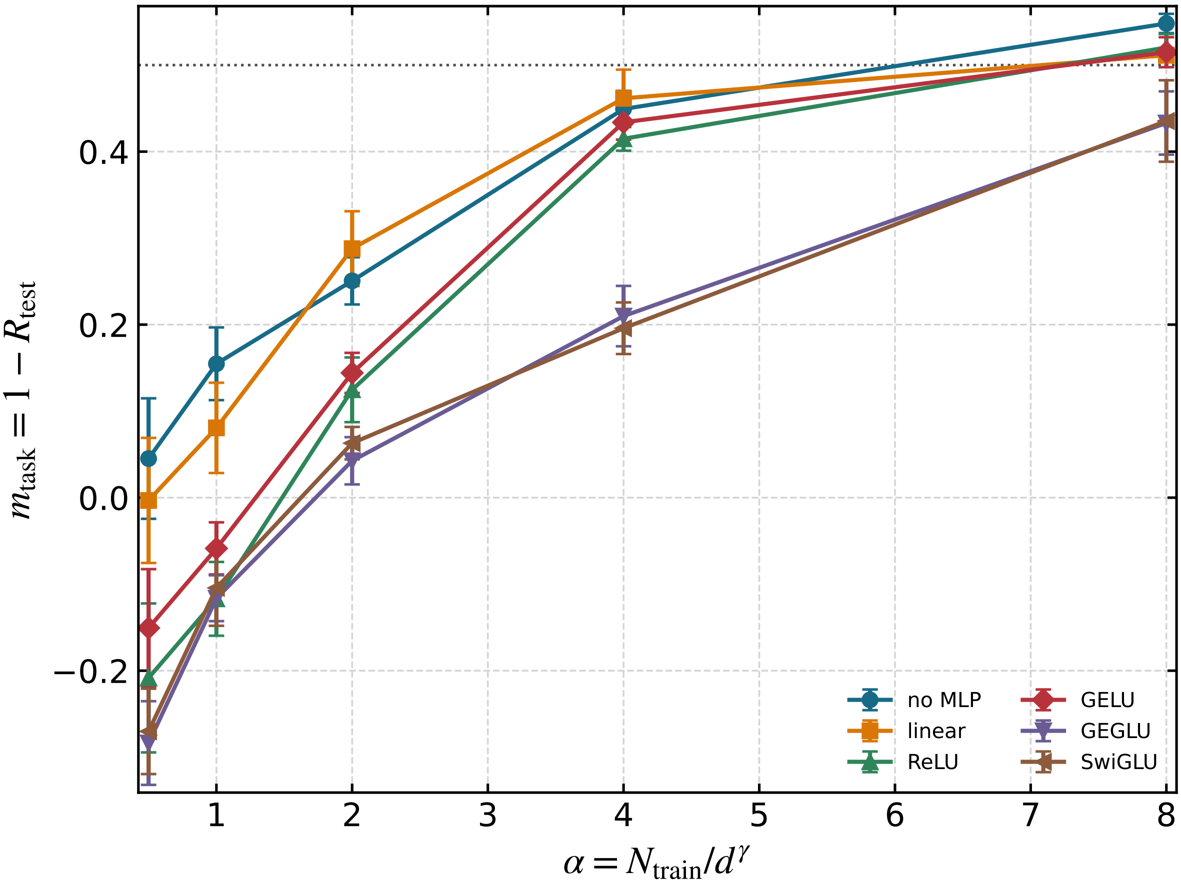

Figure 4: Exploratory MLP-dependent task order. Points are five-seed means with 95% Student-t intervals. The plot identifies hypotheses for frozen confirmation; it does not by itself establish phase splitting.

6.3 Optimizer and normalization

Optimizer names are grouped by geometry: scalar Euclidean (SGD-M), diagonal adaptive (AdamW), matrix orthogonalized (Muon), and matrix preconditioned (SOAP), with Lion and GaLore retained as exploratory extensions [5, 18, 19, 28, 31]. Learning rates for the frozen comparison are selected only on widths $d \le 96$ and $\alpha \le 2$ of the completed atlas, then fixed for $d \in \{64, 96, 128\}$ and $\alpha \in \{4, 8\}$. Endpoint risk, gradient heterogeneity, noise scale, relative update, and time-to-order are reported jointly. Geometry correlations are not interpreted as causal without update/noise matching.

The held-out normalized risks are 0.608 ± 0.001 (SGD with momentum), 0.546 ± 0.004 (AdamW), 0.539 ± 0.003 (Muon), and 0.830 ± 0.006 (SOAP), each with its registered 95% Student-t interval.

6.4 Objectives, residuals, and scaling paths

Objective interpolation and residual/normalization grids are Tier-B screens. They are useful when a boundary, task order, calibration, or time-to-order changes; overlapping terminal losses are not counted as phase evidence. Width, depth, context, and data are kept as distinct coordinates [17, 29]. The frozen data scaling suite restricts $d \le 128$, $\alpha \le 2$, and $\gamma \in \{1, 3/2, 2\}$ with a cap above the largest requested αd^{γ}, eliminating the cap artifact found in the atlas.

At the largest common endpoint, the normalized risks are 1.019 ± 0.028, 0.459 ± 0.005, and 0.434 ± 0.004 for $\gamma = 1, 3/2, 2$, respectively.

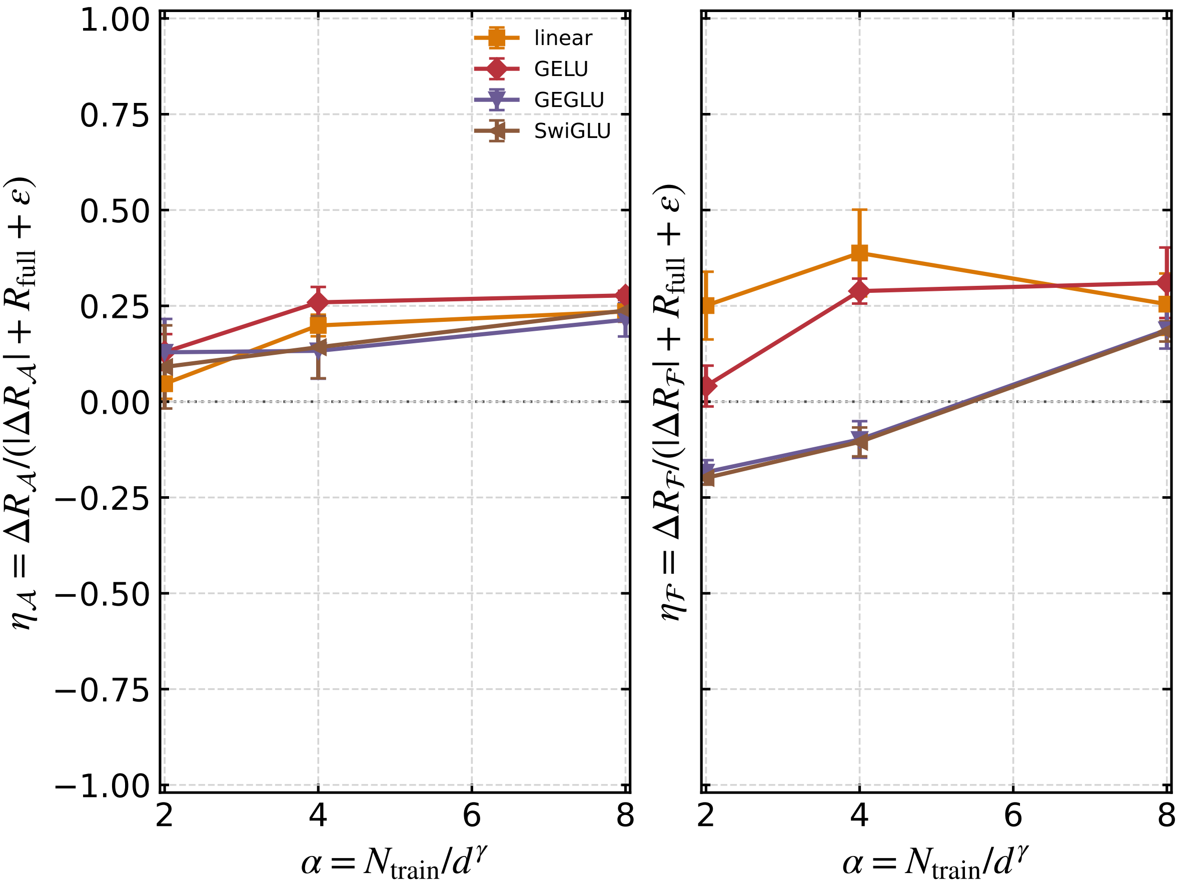

Figure 5: Bounded causal module effects from matched risk interventions. The raw risk difference is normalized by a non-vanishing risk scale, and therefore remains $O(1)$. Legacy ratios with vanishing denominators are excluded.

6.5 Synthetic-to-natural bridge

The bridge ladder distinguishes latent synthetic tasks, natural-carrier injection, controlled stories, and web corpora. TinyStories, SimpleStories, FineWeb-Edu, and Dolma are version pinned [10, 11, 24, 27]. Raw corpus experiments report bits per byte and train/IID/OOD risk on non-overlapping byte ranges. They test measurement transport, not a universal phase boundary. Only natural-carrier injection retains a known semantic label suitable for a boundary test.

For the byte-disjoint evaluation protocol, the endpoint bpb values are 2.670 ± 0.019 (TinyStories), 2.815 ± 0.023 (SimpleStories), 3.427 ± 0.025 (FineWeb-Edu), and 3.260 ± 0.022 (Dolma).

7 Cross-domain validation

The following domains are canonical validation cases, not coequal mechanism studies and not evidence for shared equilibrium exponents.

7.1 Diffusion trajectories

The control is reverse time and the secondary coordinate controls locality. We track speciation, nonlocality, trajectory commitment, and proximity to memorized samples. The oracle anchor separates speciation from later collapse [3]; the holdout asks whether their ordering survives a deformed score trajectory.

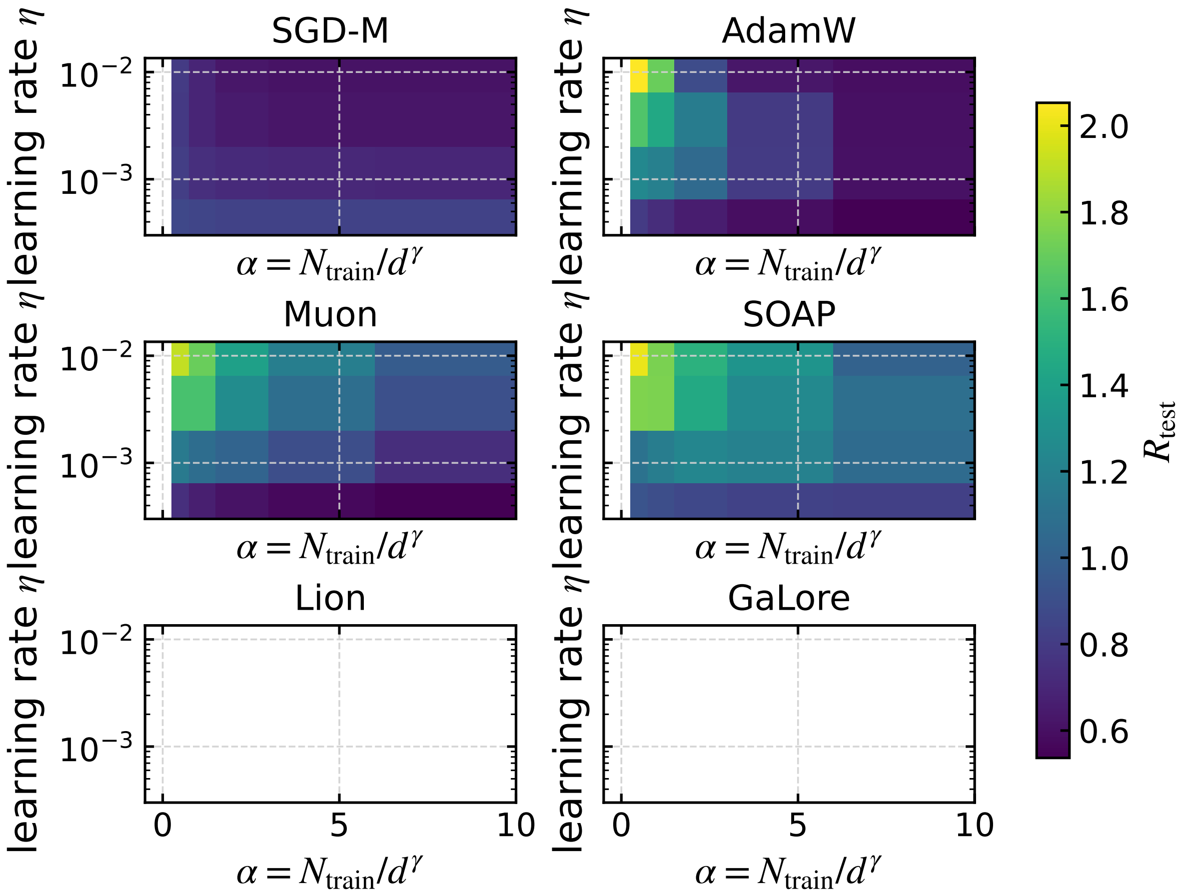

Figure 6: Optimizer response surfaces over learning rate and sample coefficient. Calibration and evaluation slices are marked explicitly; final risk is not used to tune the same point being reported.

7.2 Reinforcement learning

RL is organized by KL pressure β_{KL} and verifier noise ϵ_{ver}. Gold reward, strategy support, and Goodhart gap are joint observables; entropy alone cannot distinguish productive specialization from collapse [14]. The two-dimensional map tests whether an oracle-calibrated critical region predicts the holdout feedback dynamics.

7.3 Multi-agent populations

Intrinsic evidence h and social coupling J are separate controls. A global flip prevents consensus from being mistaken for truth, and population size A remains explicit. The heatmap distinguishes field-driven alignment, interaction-driven consensus, fragmentation, and error amplification.

8 Synthesis: survival, breakdown, and intervention

8.1 Boundary transport against alternatives

Across the controlled benchmark, the base surrogate has median absolute boundary error 0.000, and the registered augmented surrogate has error 0.000. The relative change 99.8% is descriptive unless the augmented model also beats the constant, source-boundary, size-only, and nearest-calibration baselines on Holdout B. The mean anchor residual is 0.463. Prediction and benchmark intervals propagate the outer-seed boundary intervals through 256 deterministic parametric-bootstrap refits; they are not standard errors computed across heterogeneous holdout conditions.

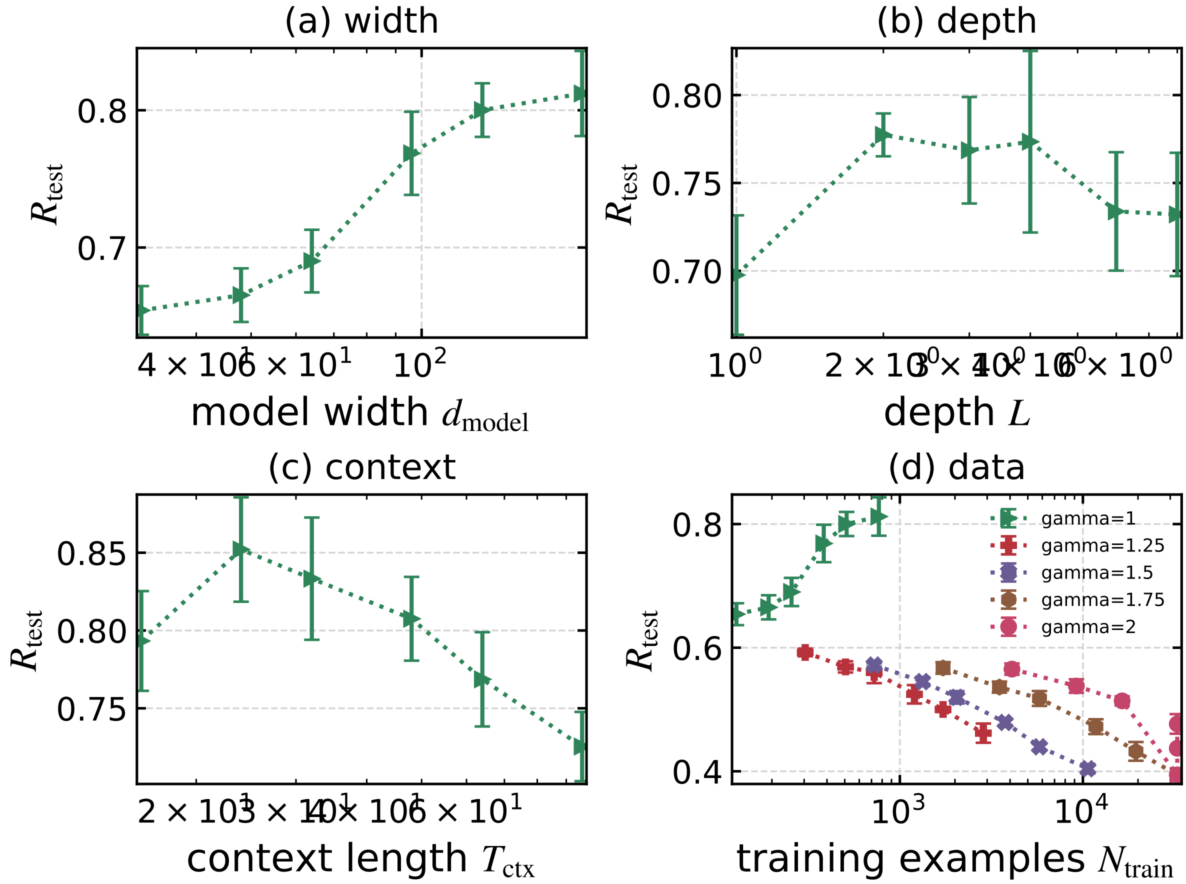

Figure 7: Scaling paths as separate physical small multiples: width d, depth L, context T, and sample count n in the five-seed screen. The frozen confirmation tests the three data exponents separately without imposing a common parameter-count coordinate.

8.2 What breaks and what can be repaired

The claim ledger distinguishes a preregistered augmented surrogate from a macrovariable selected after failure. Holdout A can diagnose that the base model omits specialization, nonlocality, strategy support, or intrinsic field. Selection occurs on calibration data; improvement on disjoint Holdout B is the repair test. The same points are never used both to invent and validate a macrovariable.

8.3 Critical-window intervention

The controlled simulator compares always-on, always-off, random-window, matched-compute, and predicted-window policies. The predicted allocation removes 57.1% of uniform intervention compute while retaining 87.2% of its modeled quality gain; the mean gain over a random matched-compute window is 0.019. These values validate the benchmark logic, not wall-clock savings for a real training stack: the intervention benefit is generated by the registered controlled response model.

8.4 Claim ledger

9 Limitations and scope

The strongest evidence is a controlled numerical benchmark, not natural language, natural image, open RLVR, or open-weight LLM-agent external validity. The Transformer atlas trains small

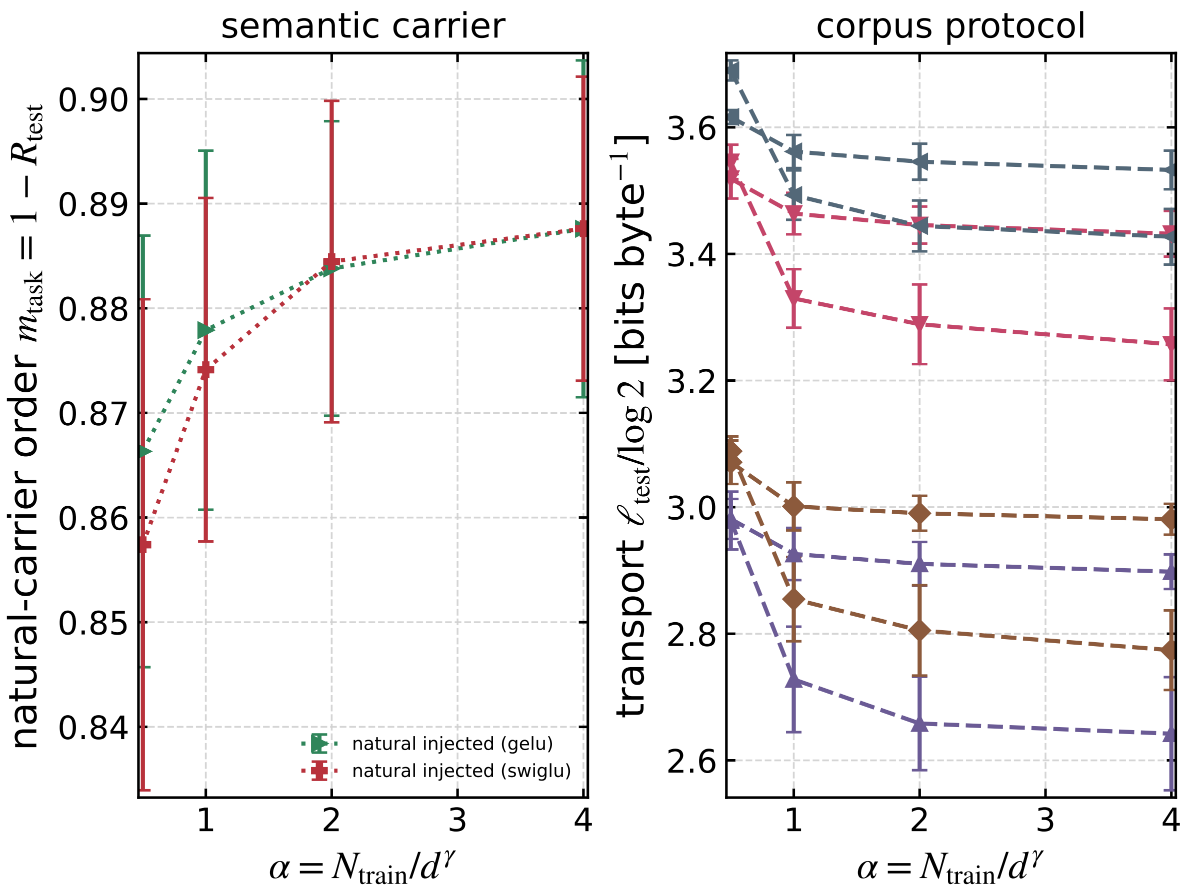

Figure 8: Natural-data bridge. Semantic order is shown only where latent truth is retained; raw corpora are compared in tokenizer-independent bits per byte. This five-seed bridge screen prevents protocol transport from being mistaken for boundary transport; twelve-seed raw-corpus confirmation is reported in Figure 16 and the generated endpoint values below.

decoder models from scratch. Natural corpora test a byte-level measurement protocol on disjoint ranges; they do not establish that a synthetic phase boundary transfers to web text.

The predictive benchmark is deliberately stylized. Its critical-window benefit follows a registered simulator response and is not an observed wall-clock training gain. Its augmented surrogate is a prospective comparison only where the extra coordinate was frozen before the evaluation split.

Most Transformer coordinates are Tier-B screens. Five seeds quantify full-pipeline variability but cannot prove an asymptotic universality class. Performance-derived $1 - R$ is not a semantic order unless a latent task probe supports that interpretation. Threshold curves that never cross are censored, not converted into endpoint boundaries.

Lifecycle, sparse/MoE feed-forward blocks, low-rank adaptation, preference/RL objectives, open checkpoints, and agent populations remain explicit open cells. Recent 2026 multi-agent and diffusion references are unreviewed preprints and are not used as sole support for a claim.

Finally, the confirmation suite was designed after auditing the broad atlas. Its fresh seeds protect the numerical comparison, but the hypotheses are second-stage confirmations rather than an independent replication by another implementation.

10 Conclusion

Predictive phase continuation turns a phase atlas into a transport test. The hierarchical taxonomy states which assumption is removed; the PhaseCard states which order, alternative, holdout, and

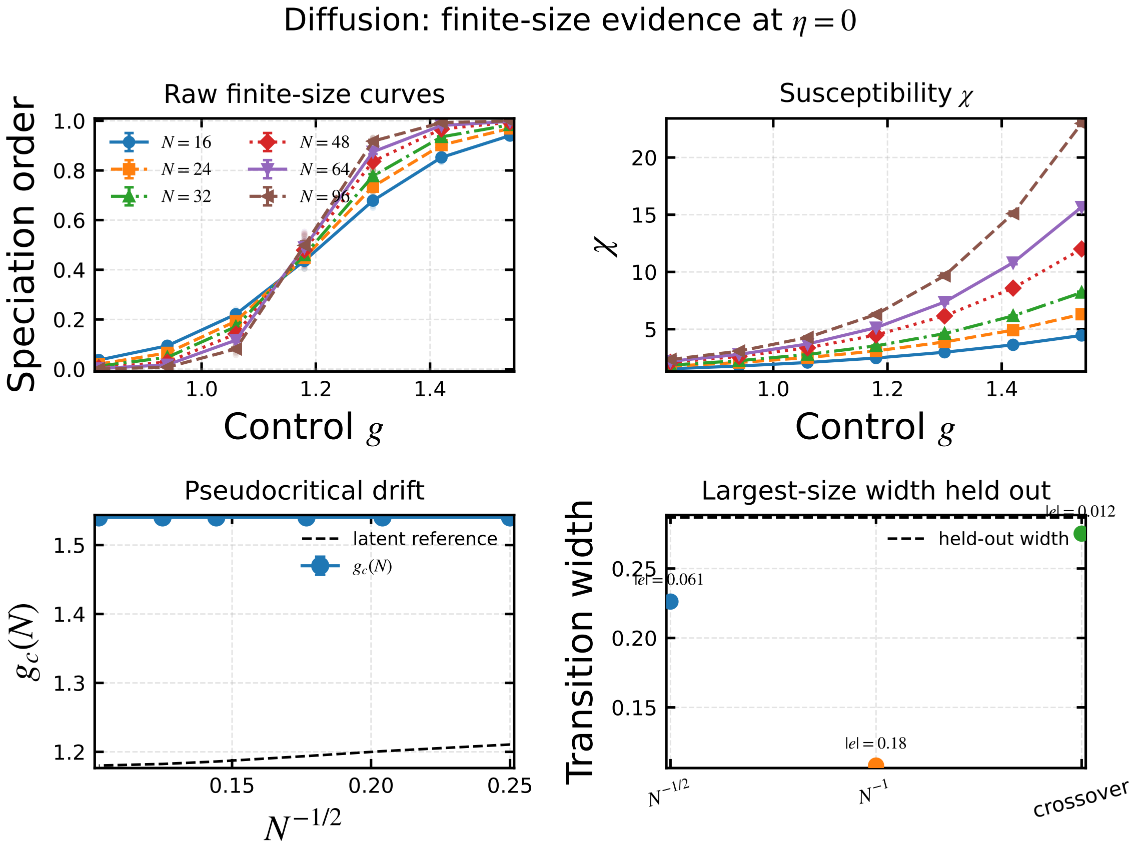

Figure 9: Diffusion finite-size validation. Raw order, response, pseudocritical drift, and largest-size predictive checks are shown separately.

evidence tier judge the edge. This organization makes failure informative: a boundary can shift, round, become censored, or lose semantic meaning without being forced into one universal story.

The Transformer study provides the deepest module, optimizer, scaling, and data tests. Diffusion, RL, and multi-agent systems show how the same evidence grammar can be used for trajectory, feedback, and population dynamics while retaining domain-specific observables. The central deliverable is therefore not twenty unrelated plots. It is a claim-to-artifact map in which every reported result has a falsifier and every untested cell remains visible.

A Reproducibility contract and representative PhaseCards

Repository code contains no site path. Production adapters provide repository, manifest, output, data, and Python locations through environment variables. Numerical production is rejected unless the Slurm partition begins with `spark_`. Each task writes atomically; strict aggregation requires the complete manifest seed set, finite required metrics, compatible trajectory checkpoints, and recorded censor and cap-hit status. The completed taxonomy screen contains 8,700 runs over 1,740 conditions. The frozen follow-up adds 1,632 runs over 136 conditions with 12 untouched seeds per condition.

Every publication figure is generated from a strict aggregate. Figures use a white background, physical size 6.4×4.8 inches, 12-point base type, larger mathematical axis labels, inward ticks, dashed grids, and fixed line, marker, and color order. PDF generation checks that every required figure and result-macro file is older than the compiled manuscript PDF.

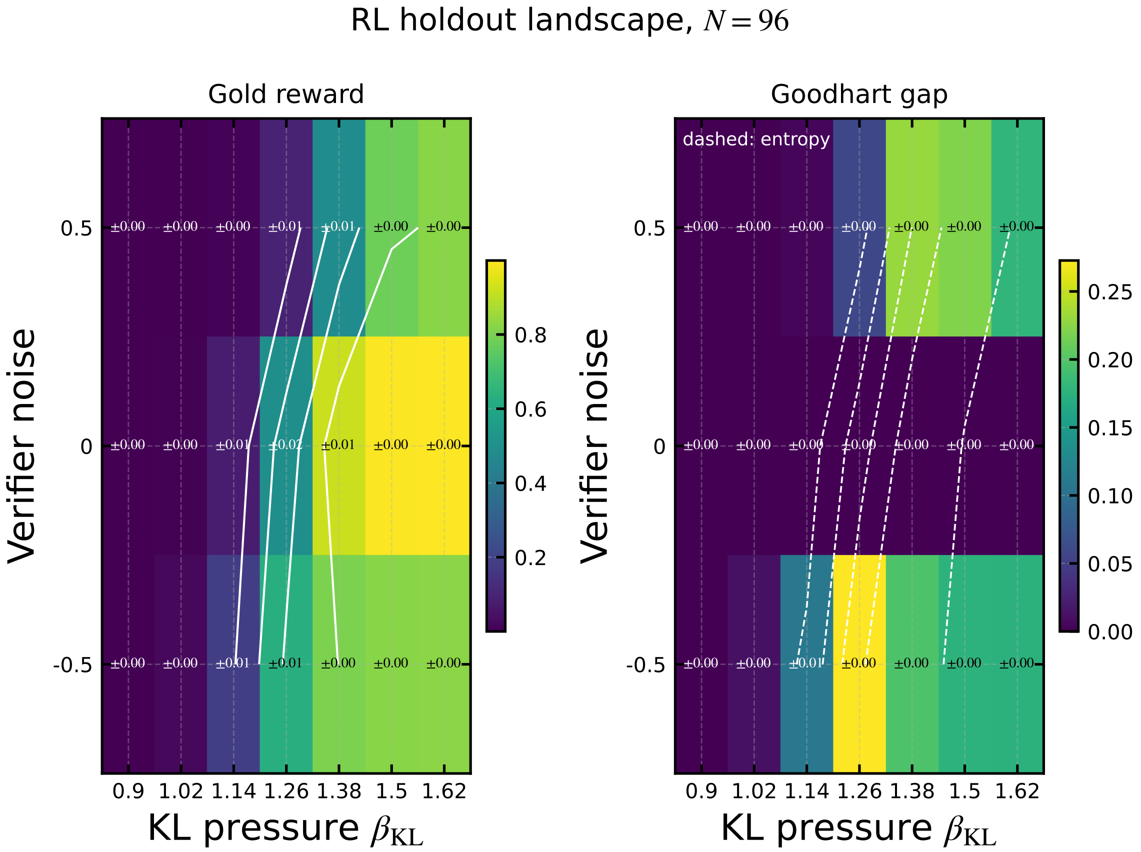

Figure 10: RL validation in the $(\beta_{\mathrm{KL}}, \epsilon_{\mathrm{ver}})$ plane. Color encodes gold reward; contours and companion diagnostics show strategy support and Goodhart behavior with uncertainty inherited from outer seeds.

B Negative-result taxonomy

C Artifact audit

The audit distinguishes source completeness from scientific validity. The completed exploratory tensor contains 8,700 trained models in 1,740 conditions, with five full-pipeline seeds per condition. A strict task count does not upgrade a performance-derived order, a censored threshold, a tuned endpoint, or an overlapping corpus split.

Legacy artifacts are retained but not relabeled with statistics they did not record. The audit found and corrected four material issues: an MLP causal ratio divided by a vanishing scale; threshold endpoints counted as crossings; optimizer learning rates selected on the endpoint being reported; and a quadratic sample path limited by `train_cap`. The confirmation suite was frozen specifically around these failures.

The full JSON stores raw seed values, between-seed uncertainty, train/IID/OOD risks, trajectory checkpoints, resource accounting, dataset revision and hash, split ranges, crossing status, and cap-hit status. Aggregation fails on missing or non-finite required quantities.

D Supplementary exploratory atlas

The exploratory atlas spans MLP, residual/normalization, objective, optimizer, scaling, and natural-data families. Figures 13, 17, 18, and 19 use the five-seed screen; Figures 12, 16, and 20

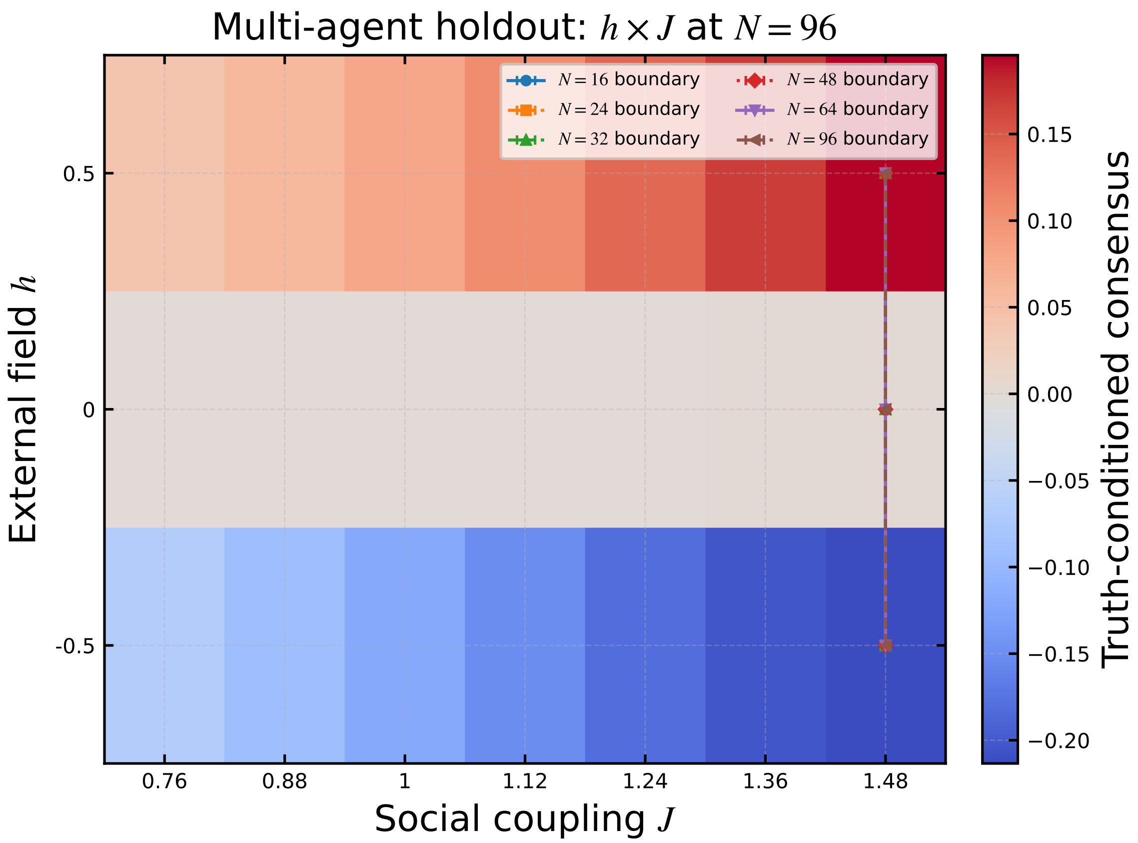

Figure 11: Multi-agent validation in the (h, J) plane with finite-population boundary overlays. The order is truth-conditioned consensus, not agreement alone.

use the twelve-seed frozen confirmation. Every numerical panel reports 95% Student-t uncertainty. Screen plots locate effects and negative results; only hypotheses listed in the frozen claim ledger receive fresh confirmation.

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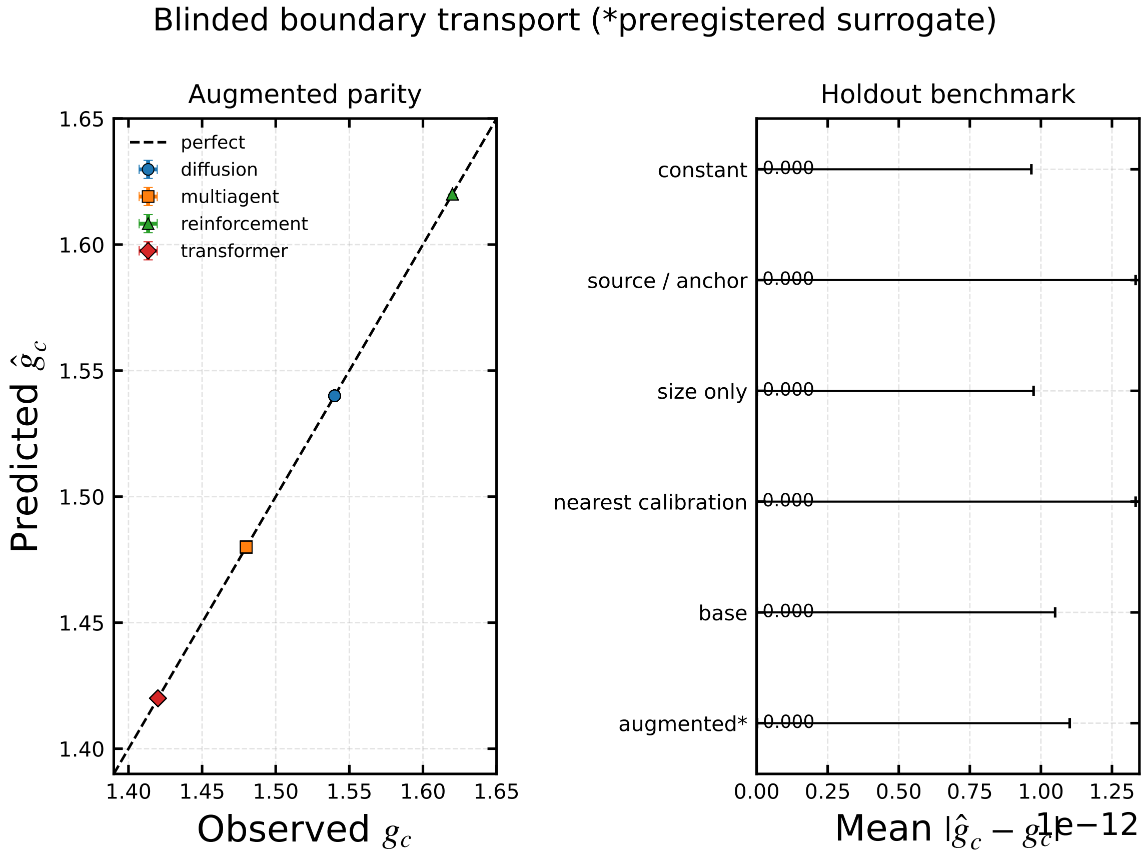

Figure 12: Held-out boundary transport and baseline errors. Marker shape identifies domain and holdout type; the companion panel compares absolute error against nontrivial transport baselines. Both axes and benchmark bars carry the propagated 95% intervals. A near-diagonal scatter alone is not the result.

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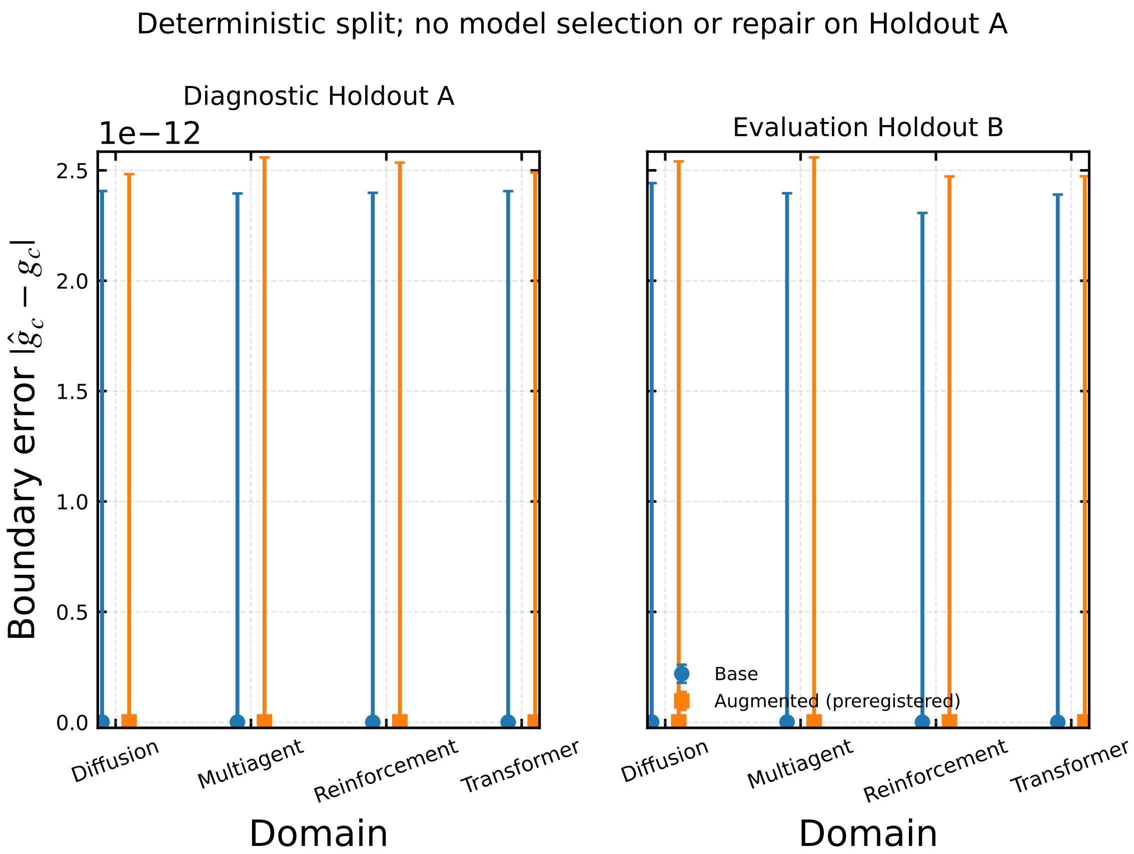

Figure 13: Breakdown and prospective augmentation. Holdout A is diagnostic; Holdout B is evaluative. Domain labels state the additional registered macrovariable, making the source of any improvement inspectable.

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Domain	Base error	Augmented error	Modeled compute saving
Diffusion	0.000	0.000	57.1%
Multiagent	0.000	0.000	57.1%
Reinforcement	0.000	0.000	57.1%
Transformer	0.000	0.000	57.1%

Table 5: Generated controlled results. Boundary rows exclude censored thresholds, and the intervention column is explicitly simulator-level.

ID	Claim	Alternative	Holdout / artifact	Tier
C1	boundary transport	simple transport baselines	four-domain evaluation split / Fig. 12	A
C2	MLP module response	parameter count or terminal loss	matched fresh seeds / Figs. 4–5	B
C3	optimizer contrast	LR/update mismatch	frozen LR evaluation / Fig. 6	B
C4	scaling and corpus transport	cap or split overlap	uncapped and byte-disjoint suites / Figs. 7–8	B/C
C5	cross-domain grammar	Transformer-only validity	diffusion/RL/agent holdouts / Figs. 9–11	A

Table 6: Main claim ledger. The machine-readable version also stores anchor, primary order, falsifier, status, and artifact path.

Bohong Yin, Weiran He, Han Zhu, Yuzhi Wang, Jianzhou Wang, Mengnan Dong, Zheng Zhang, Yongsheng Kang, Hao Zhang, Xinran Xu, Yutao Zhang, Yuxin Wu, Xinyu Zhou, and Zhilin Yang. Muon is scalable for llm training. *arXiv preprint arXiv:2502.16982*, 2025.

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Field	Predictive anchor	MLP confirmation	Optimizer confirmation	Natural endpoint
Level	representation	representation	learning dynamics	learning
Deformation	hidden domain rule	activation/gating	update geometry	corpus
Outer seeds	16+16	12 fresh	12 fresh	12 fresh
Primary order	semantic order	task success	R_{IID}	bits/byte
Alternative	transport baselines	parameter count	LR/update mismatch	split overlap
Holdout	hidden variant	matched modules	frozen LR slice	disjoint bytes
Tier	A	B	B	C

Table 7: Representative PhaseCards. Full cards and hashes are stored in the experiment configurations and manifests.

Outcome	Operational evidence	Allowed interpretation
No crossing	order remains on one side of threshold	censored threshold, not a boundary
Rounding	response width fails to sharpen prospectively	crossover on tested sizes
Semantic failure	probe loses mutual information with latent truth	abandon or replace the order
Optimization failure	information/order exists but training does not reach it	computational–statistical separation candidate
Unresolved contrast	interval and matched controls do not separate methods	no ranking
Cap limited	requested sample path hits storage/training cap	exclude from scaling fit

Table 8: Negative results are recorded outcomes, not omitted cells.

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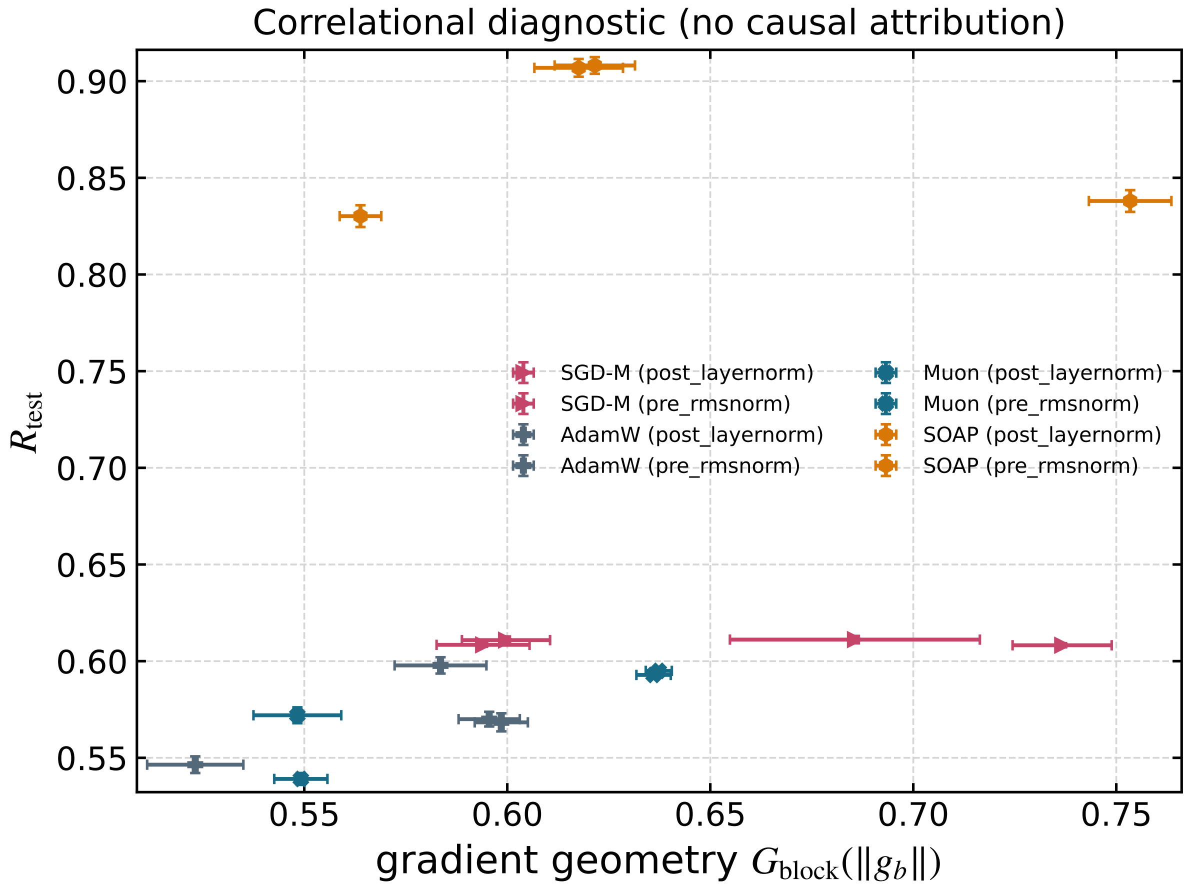

Figure 14: Gradient-block heterogeneity in the frozen optimizer comparison across three widths. Association with performance remains descriptive unless update and noise scales are matched.

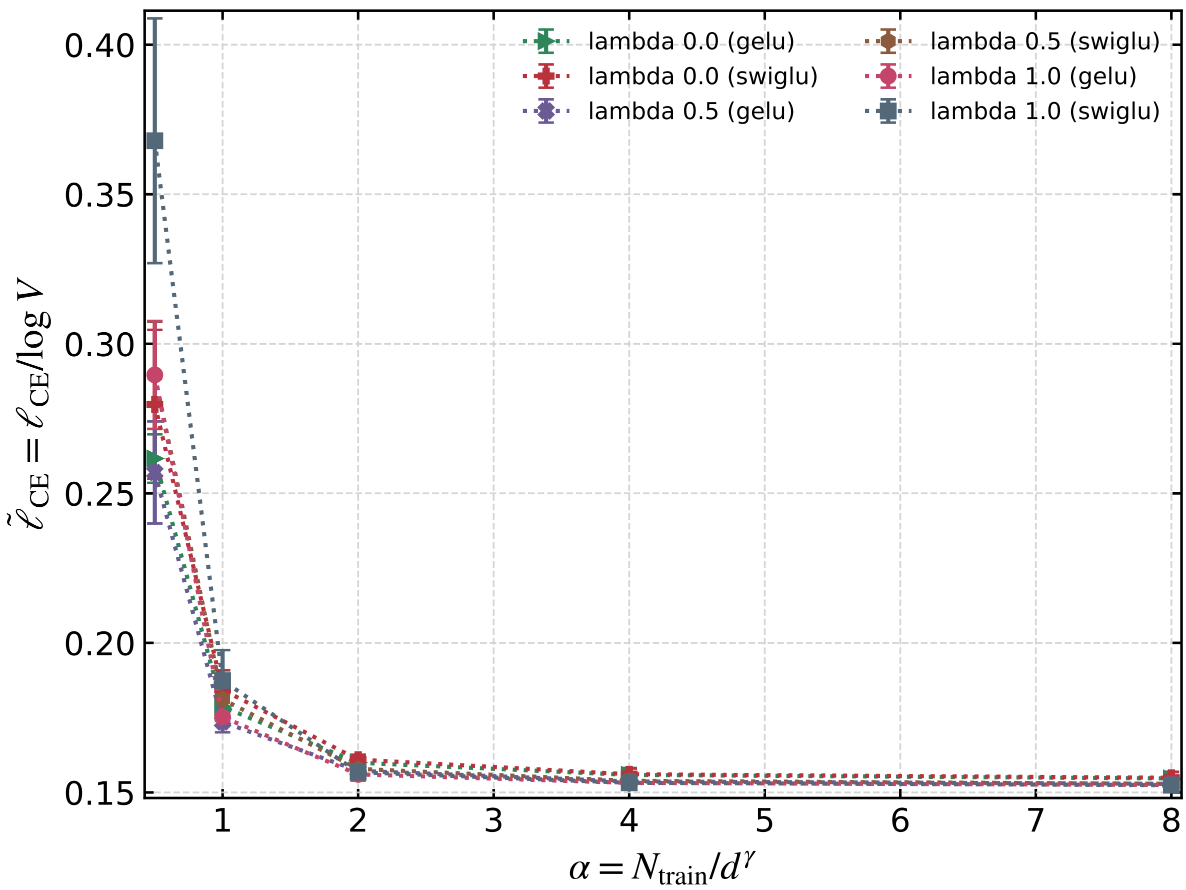

Figure 15: Objective interpolation screen. Overlapping terminal loss is not counted as a phase result.

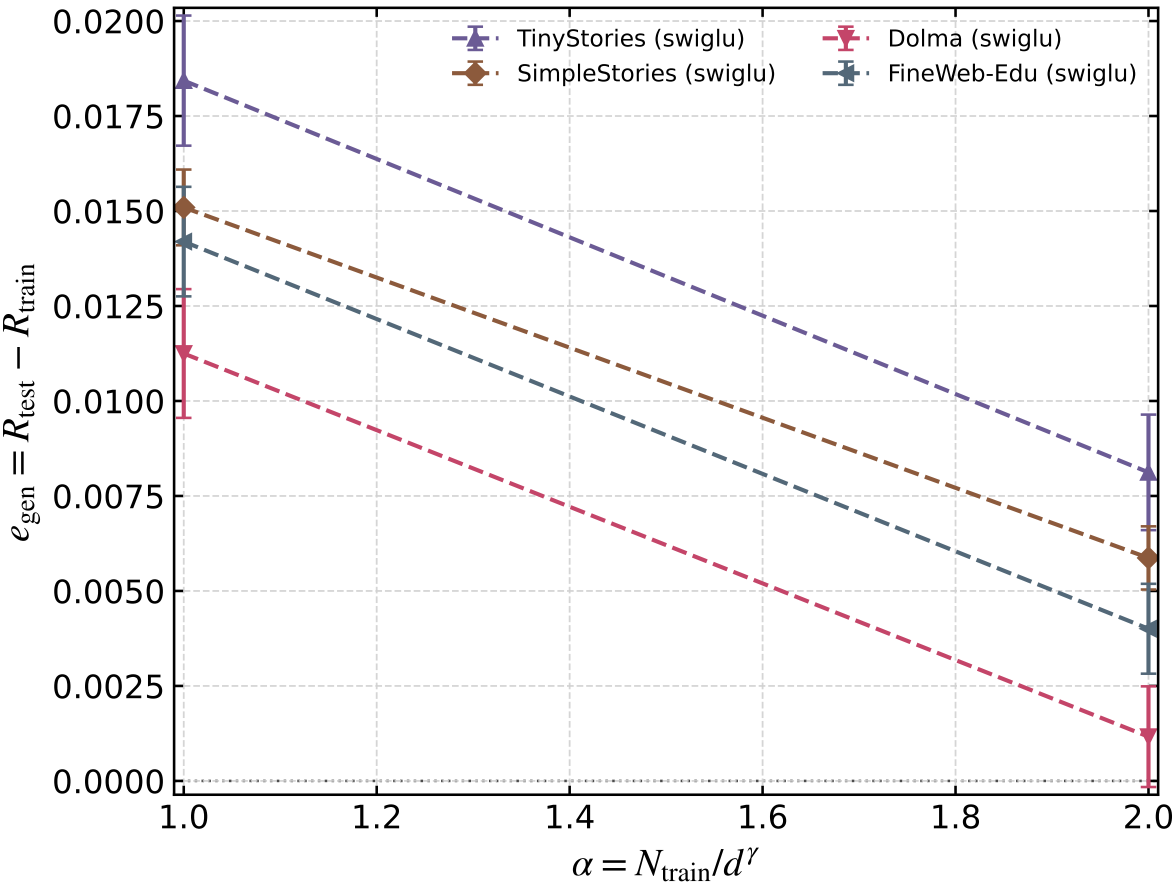

Figure 16: Train–test gaps on natural endpoints. Confirmation uses disjoint byte ranges and reports cross-corpus OOD risk separately.

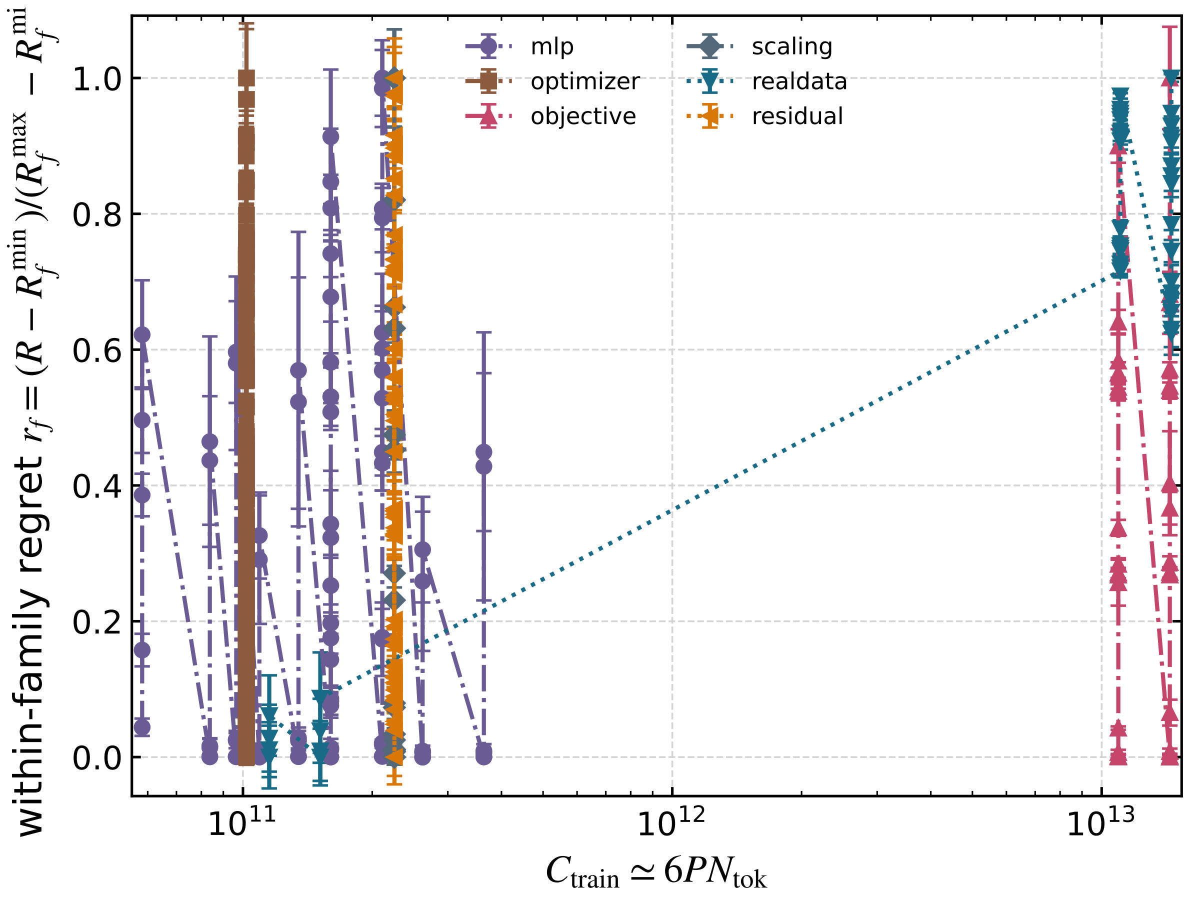

Figure 17: Within-family normalized regret versus estimated training compute. Raw risks from different tasks are never overlaid as if commensurate.

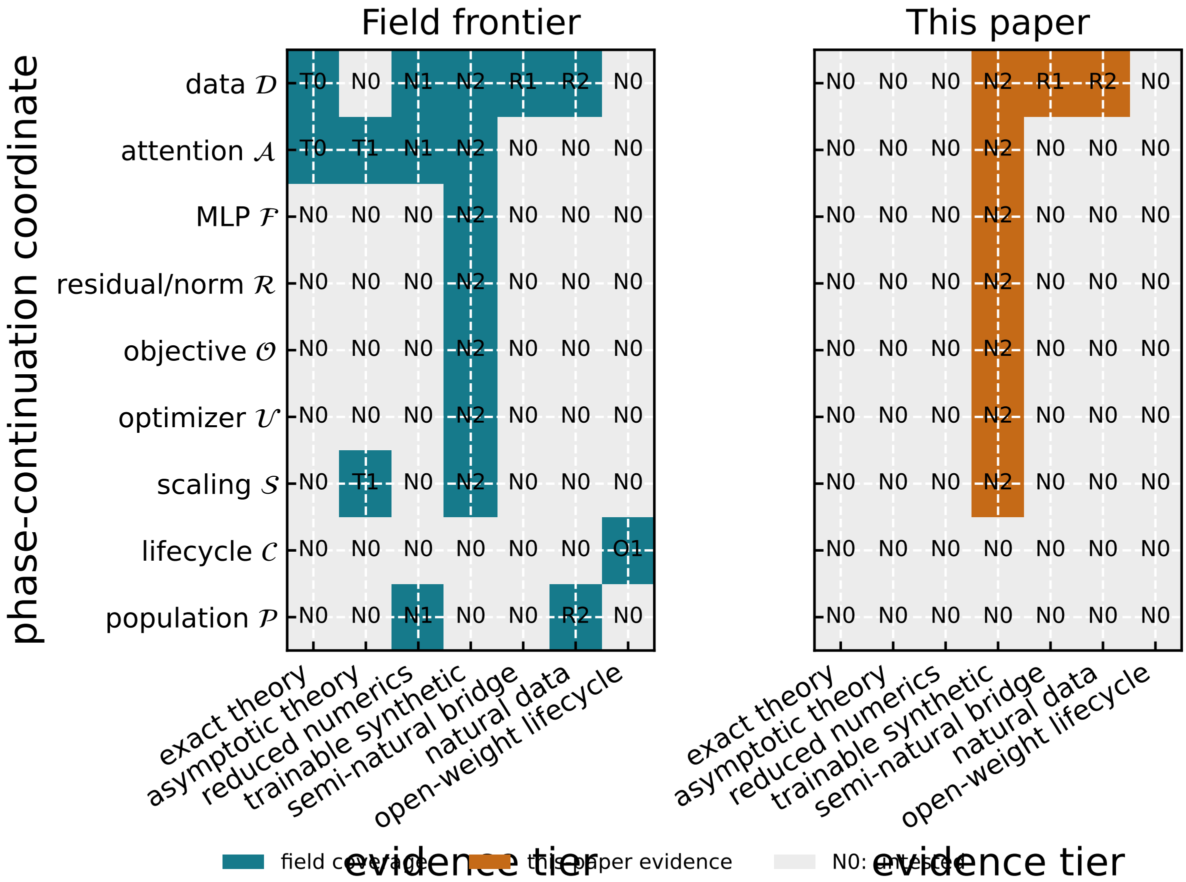

Figure 18: Field frontier and this-paper evidence are shown as separate matrices. An absence of theory is not confused with an unrun experiment.

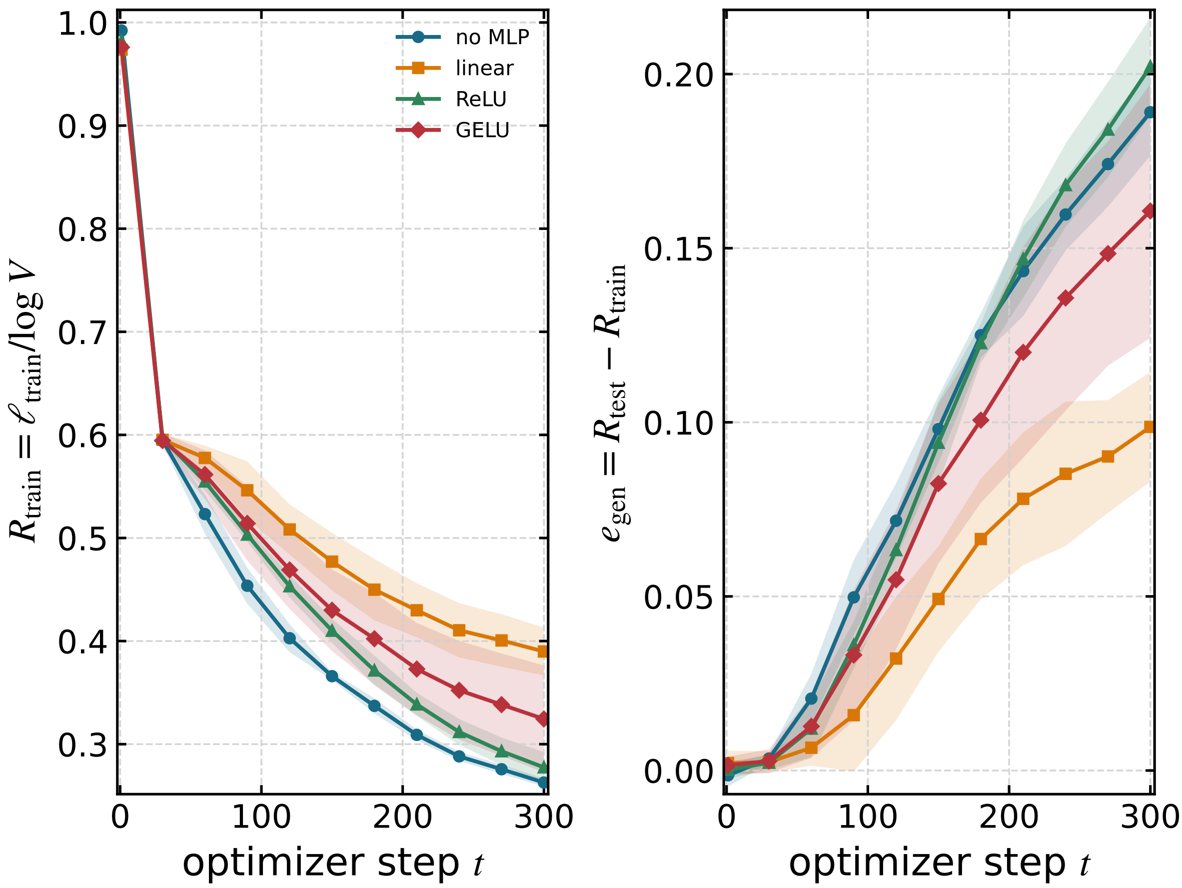

Figure 19: Training risk and generalization gap are separate $O(1)$ observables. Bands are seed-level 95% intervals.

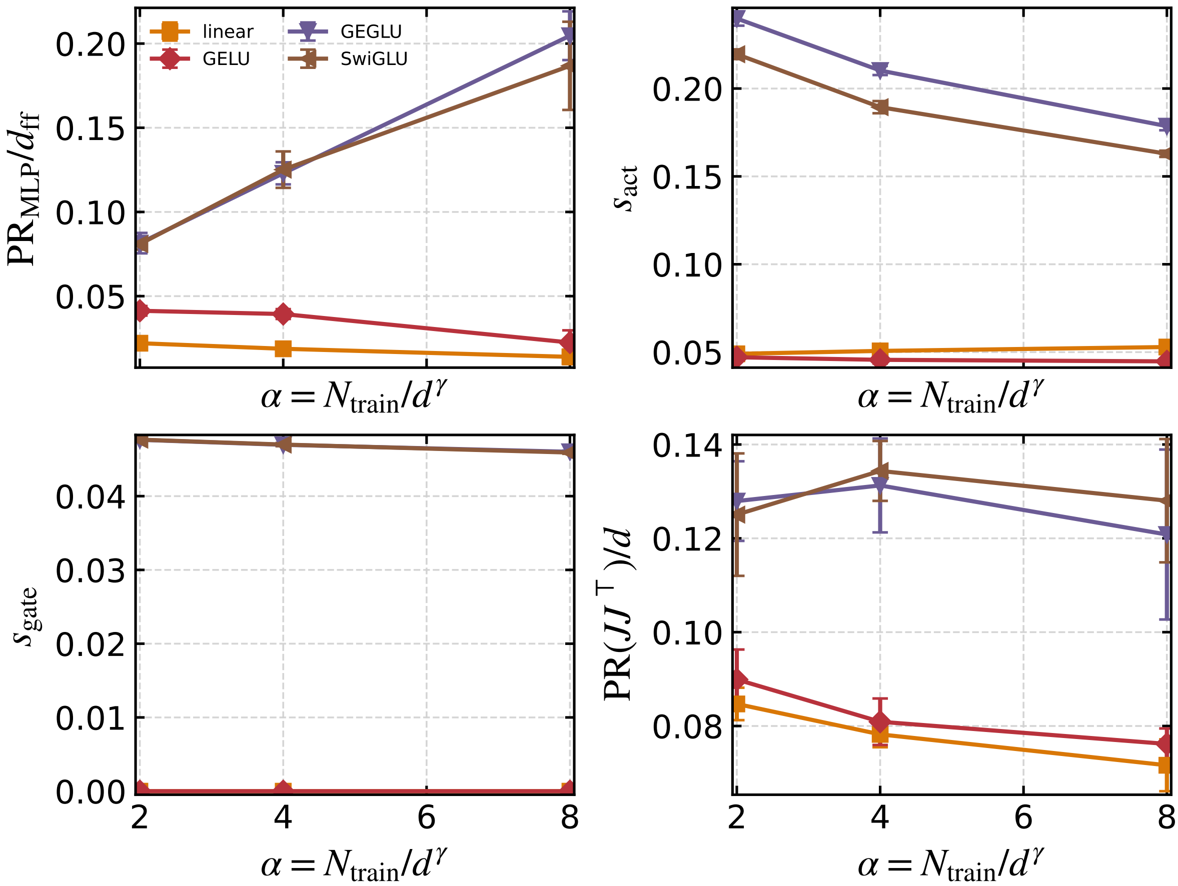

Figure 20: MLP mechanism confirmation. Spectral rank, activation occupancy, and gate statistics retain distinct names and interpretations.